Primitive streams · my notes L12

numbers travel without boxes

Boxed vs primitive 📦

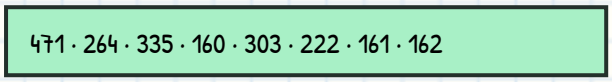
Stream<Integer>: every number in its own box



… objects, garbage

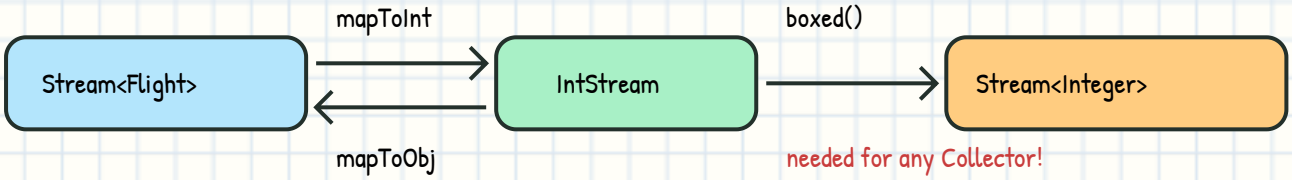
1000 == 1000 → false!
(cache: -128..127)

IntStream: plain ints on a belt



sum() → 2078
summaryStatistics() → 8 · 2078 · 160 · 471 · 259.75

Getting in and out 🔁



needed for any Collector!

range vs rangeClosed 📏

`range(1, 5)` → 1 2 3 4 = **10**

`rangeClosed(1, 5)` → 1..5 = **15**

`iterate(1, r -> r <= 42, r -> r + 10)` [9]
→ [1, 11, 21, 31, 41]

Empty stream? 🫙

`sum()` → 0

`average()` → OptionalDouble.empty

`getAsDouble()` → NoSuchElementException

use orElse(..)

One pass, 5 answers 📊

```
mapToInt(f -> f.seats() - f.booked())
.summaryStatistics()
```

empty seats: min=2 max=63 avg=33.75

toString uses your locale → Locale.ROOT

Zip with index 🔢

```
range(0, n).boxed()
.collect(toMap(i -> i + 1, ..))
```

→ {1=SQ318, 2=SQ12, …, 8=GA827}

Money is not a double 💰

double ✗

0.1 + 0.2 = 0.30000000000000004

10 × 0.1 (loop) = 0.9999999999999999

0.1+0.2+0.3 = 0.6000000000000001

0.3+0.2+0.1 = 0.6 ← order matters!

BigDecimal ✓

reduce(ZERO, add) → 4429.84

divide(n, 2, HALF_UP) → 369.15

long cents: 442984 → 4429.84

10 × 0.10 = 1.00 exactly

chars() 🔤

`"SIN".chars()` → [83, 73, 78] ints!

`mapToObj(c -> (char) c)` → [S, I, N]

flight codes: '8' appears 5 times

"Pay 💳": chars=6, codePoints=5

Random streams 🎲

`new Random(42).ints(6, 1, 7)`

→ [3, 4, 1, 3, 1, 2], every run

tests: seed it · secrets: SecureRandom

RandomGenerator [17]

Worked example: flight load report

BEFORE: List<Integer> + MAX_VALUE loop · AFTER [8]: mapToInt + summaryStatistics · MODERN: record FlightLoad with exact BigDecimal % and a cross-multiplying comparator · SQ21 98.76% … MH602 61.25% · TOTAL 87.01% · all 3 agree ✓

Exercise: seats + PNRs ✈

occupancy % per flight {GA827=74.07, …, SQ21=98.76} · ≥ 90%: [SQ21, SQ318, SQ938, TR808] · PNR chars: Q=5 is the most frequent · 16 digits in 8 PNRs

★ Remember

primitives = no boxes · range is half-open · empty average is Optional · box before a Collector · money: BigDecimal or cents · chars() are ints · next → L13 advanced operations